Case Study Report

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Introduction

In this case study, I looked at simulations of tiny particles, such as water droplets in clouds for example, moving through turbulent air. Understanding how they form clusters can help explain things like how rain forms. The data set has 89 simulation runs with three input variables, that includes the Reynolds number (Re) which measures how turbulent the flow is, the Froude number (Fr) which tells how strong gravity is, and the Stokes number (St) which measures how much inertia the particles have. And for each run, I looked at four summary statistics, that includes the mean which is the average cluster volume, the variance which measures how much the cluster volumes differ from the average, the skewness which describes whether the distribution is more spread out toward smaller or larger clusters, and the kurtosis which tells us how likely the distribution is to contain unusually large clusters. For this case study, I mainly interested in inference, meaning which inputs change the cluster volumes, how big these effects are, whether they are linear or not, and whether they are all the same for all four summary statistics along with predictions for new combinations of Re, Fr, and St.

A fair question is why scientists may need a machine learning model when they already have a scientific model, which is the simulation itself. Well these simulations are very expensive, which is why there are only 89 runs for the training data, and that there is no simple formula for how the summary statistics depend on Re, Fr, and St together, so this relationship has to be learned from the data. The raw simulation results alone are also insufficient since each setting was tested only once, so it makes it difficult to isolate the effects of individual input variables. So by using information from all 89 runs, our machine learning model can better identify these relationships and can provide more accurate predictions.

Since inference is one of my main goals, I used interpretable log-scale linear models where each term helps explain the effects of the input variables through confidence intervals and hypothesis tests. With only 89 runs, the data cannot reliably support every possible model term, so variable selection is important for identifying which input variables have the greatest influence on the results. To address this, I evaluated the log transformation using Box-Cox, used natural splines to capture nonlinear relationships with St and retained only the interaction terms that were statistically significant and improved cross-validation performance.

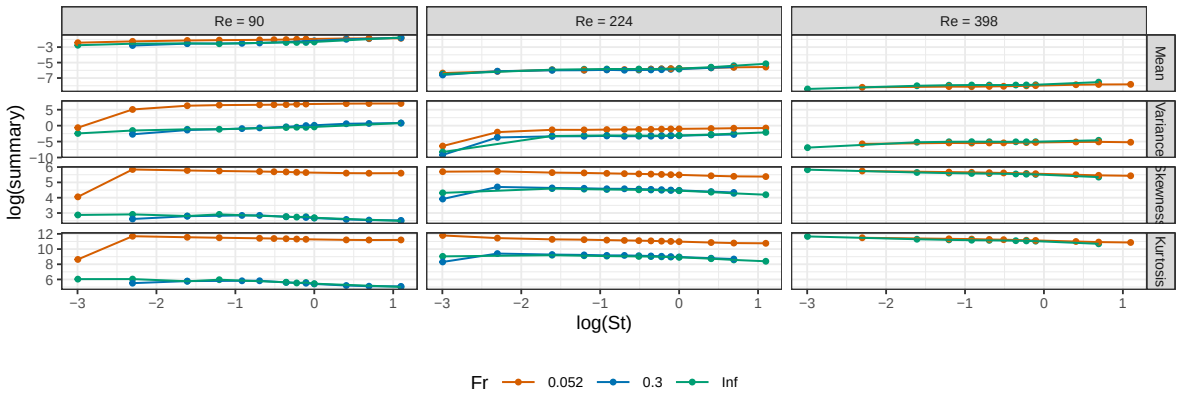
Methodology

Preparing the data

The data contains the first four raw moments of the cluster volumes in which I used to calculate the mean, variance, skewness, and kurtosis. All four summary statistics have a huge range, such as the variance ranges from about 10^{-4} to 10^{3}, thus I need to transform them. To pick a transformation, I sued the Box-Cox method, which just finds the power λ that best fits the data where $\lambda = 0$ means taking the log. For all four summary statistics, the best λ was very close to 0 and the log transformation also had a lower CV error than the other options that I tried. So I took the log of each summary statistic which also makes each effect easy to read as how many times larger or smaller a summary statistics has become. Re and Fr only have three values each, so I treated them as categorical, while St has 14 values bunched near 0, so I used $\log(St)$. One limitation of doing this method is that categorical Re and Fr mean my models can only predict and make inference at the simulated values, not for values that were not tested.

What I saw in the data

The figure below showcases the result from all 89 simulation runs for each of the four summary statistics. Each line connects runs with the same Fr level, which allows me to see how St affects the results while Re and Fr remain constant. The figure showed me three important things about the data. The first observation is that the relationship with St is not linear even after taking $\log(St)$. The second observation is that Re and Fr seem to interact. It is seen that strong gravity ($Fr = 0.052$) greatly increases the variance, skewness, and kurtosis at $Re = 90$, but it increases less at $Re = 224$ and it barely increases at $Re = 398$. The third observation is that the lines for $Fr = 0.3$ and $Fr = \infty$ are almost identical.



Models I tried

I started with a simple linear model and then I kept increasing its complexity, fitting each model separately for each summary statistic. Model 1 is linear in St, Re, and gravity, and model 2 is similar but uses $\log(St)$ instead. Model 3 uses a cubic polynomial in $\log(St)$. Model 4 adds an Re $\times$ gravity interaction. After that, I tested whether I could replace Fr with a simpler gravity variable that has two categories, which are strong gravity ($Fr = 0.052$) and weak or no gravity ($Fr = 0.3$ or ∞). This gave almost the same CV error, and F-tests found no difference between $Fr = 0.3$ and no gravity for the variance, skewness, and kurtosis ($p > 0.5$). For the mean, weak graivty did make a small difference of about 2% to 10% that was statisticall significant, but the CV error barely changed, so I kept the simpler gravity variable for all four summary statistics. From there, model 5 adds an St $\times$ gravity interaction to model 4 and model 6 adds an St $\times$ Re interaction instead. Model 7 fits a separate St curve for each of the 6 Re-gravity combinations, which has 24 coefficients for only 89 runs. Model 8 and model 9 are lasso and ridge versions of model 7. Ridge reduces the size of all coefficients to make the model more stable, while Lasso can shrink some coefficients to zero, helping identify which variables are most important. For both methods, I used cross-validation to determine the appropriate amount of regularization.

Picking the best model

I compared all the models with 10-fold cross-validation, which measures RMSE on the log scale, so an RMSE of 0.1 means the predictions are usually off by roughly 10%. Since I wanted my models to be interpretable, I focused on selecting simpler models that still performed well in cross-validation. I also repeated the model selection process using 20 different random splits to make sure that the results were consistent and not due to chance. For the interactions, I also used nested F-tests and only kept an interaction if it was statistically significant and it also lowered the CV error. Lasso also kept all the main effects and the interaction terms between Re and gravity do all four summary statistics, supporting my model selection decisions. I initially used polynomial regression to capture the nonlinear relationship with St, but polynomials can produce sharp

curves near the edges of the data. This was especially concerning since my noisiest runs ($St = 0.05$) and four test points were located near these boundaries. To address this, I tested natural cubic splines and smoothing splines. All three methods had similar cross-validation errors for the mean (0.074 for polynomials, 0.070 for natural splines, and 0.075 for smoothing splines). However, polynomials have shown sharp changes outside the observed data while natural splines maintained a straight-line pattern beyond the boundaries. So I chose natural splines since they provided a more stable and reliable approach near the edges of the data.

Final Model

The final models that I selected were model 4 for the variance, skewness, and kurtosis, and model 5 for the mean, in which both using a natural spline in $\log(St)$. Let $x = \log(St)$. Let $R_{224} = 1$ if $Re = 224$ and $R_{398} = 1$ if $Re = 398$ (both would be 0 if Re = 90). And let $G = 1$ for strong gravity while $G = 0$ is for weak or no gravity. Then model 4 is

$$\log(\text{summary}) = \underbrace{f(x)}_{\text{effect of St}} + \underbrace{b_{224}R_{224} + b_{398}R_{398}}_{\text{effect of Re}} + \underbrace{b_G G}_{\text{effect of gravity}} + \underbrace{b_{224,G}\,R_{224}G + b_{398,G}\,R_{398}G}_{\text{Re}\times\text{gravity}} + \varepsilon$$

where $f(x)$ here is the natural spline curve for St, b_{224} and b_{398} shift this curve for $Re = 224$ or 398, b_G shifts the curve for strong gravity, and $b_{224,G}$ and $b_{398,G}$ allow the effect of gravity to be different at each Re level. Model 5 for the mean also let $f(x)$ have a different shape under strong gravity. The degrees of freedom of $f(x)$ is chosen by the lowest CV error, which are 4, 6, 3, and 2 for the mean, variance, skewness, and kurtosis.

Results

	Mean	Variance	Skewness	Kurtosis
M1: linear	0.167	1.899	0.601	1.161
M2: log(St)	0.136	1.724	0.608	1.174
M3: cubic in log(St)	0.133	1.658	0.610	1.183
M4: + Re x Fr	0.098	0.798	0.229	0.410
M4 with gravity variable	0.101	0.791	0.229	0.410
M5: + St x gravity	0.074	0.788	0.277	0.511
M6: + St x Re	0.129	0.838	0.271	0.492
M7: separate curves	0.148	0.684	0.231	0.405
M8: lasso	0.091	0.939	0.239	0.411
M9: ridge	0.377	0.971	0.465	0.893
Final model (natural spline)	0.069	0.783	0.226	0.390

The table above shows the CV error for each model. I noticed that models 2 and 3 does help a little, but the big improvement comes from model 4's Re × Fr interaction which cuts the CV error by more than half for th variance, skewness, and kurtosis. For the mean, model 5 does even better, which lowered the CV error from 0.101 to 0.074. Adding St × Re in model 6 makes every summary statistic worse. I also need to consider the bias-variance tradeoff where models 1 through 3 are too simple which results in high bias, while model 7 does poorly for the mean, which is a sign of high variance. Ridge does poorly since it shrinks the large Re and gravity effects along with the small ones. Based on the model selection results, Model 5 was selected for the mean while Model 4 was selected for the other three summary statistics. These choices were also consistent across 20 random splits, with Model 5 selected 19 out of 20 times for the mean and Model 4 was selected for all 20 times for the other three statistics.

The interactions in the final models are also statistically significant. I used an F-test to compare each final model to the same model without the interaction. What I found is that the Re × gravity interaction was highly significant for all four summary statistics and so was the St × gravity for the mean. Both interactions stayed significant even after removing the noisy runs at $St = 0.05$. For uncertainty, predictions are typically off by a factor of $e^{\text{CV error}}$, which is about 7% for the mean, 25% for the skewness, 48% for the kurtosis, and a factor of 2.2 for the variance.

How Re, gravity, and St affects the clusters

	Mean	Variance	Skewness	Kurtosis
Re 90 to 224	0.03 (0.029, 0.031)	0.055 (0.038, 0.081)	5.7 (5, 6.3)	29 (24, 36)
Strong gravity, Re 90	1.4 (1.3, 1.5)	750 (470, 1,200)	17 (15, 19)	260 (210, 340)
Strong gravity, Re 224	1 (0.97, 1.1)	7.5 (4.9, 11)	3 (2.6, 3.3)	8.7 (6.9, 11)
Strong gravity, Re 398	0.87 (0.81, 0.93)	0.56 (0.32, 1)	1 (0.86, 1.2)	1.1 (0.8, 1.5)

The table above compares two settings at a time, where a setting is just one combination of Re, Fr, and St, and it shows how many times larger or smaller each summary statistic becomes alongside with a 95% confidence interval. In each row, only one input changes. Unless stated otherwise, the comparisons are made at $St = 0.5$ with no gravity. The results show that Re has the greatest effect. Increasing the Re from 90 to 224 would make the mean about 30 times smaller while the skewness becomes about 5.7 times larger. What this suggest is that stronger turbulence generally produces smaller clusters, although a few clusters can still be much larger than the others. Another thing to note is that strong gravity does have a greater effect whenever the turbulence is weaker. Like from the table, strong gravity increases the variance about 765 times at $Re = 90$, only about 7 times at $Re = 224$, and lowers the variance to about half at $Re = 398$. This could mean that gravity causes particles to settle and form larger clusters while stronger turbulence breaks these clusters apart. I see that gravity changes the mean by at most about 40%, so gravity mainly changes how much cluster sizes differ rather than the average cluster volume.

Conclusion

So using cross-validation, model 4 was the best simpler model for the variance, skewness, and kurtosis, while model 5 was the best for the mean since it accounts for how the relationship between St and the mean cluster volume changes with the strength of gravity. These models provide accurate predictions while staying simple enough to understand how each input variable affects the cluster volumes. However, one limitation is that some test points have St values outside the training range, so those predictions are less reliable. My main finding is that Re has the strongest effect on cluster volumes, which shrinks the average cluster volume but it raises the skewness and kurtosis. Another finding is that gravity only matters when it is strong, and its effect fades as turbulence grows. Also that gravity changes the average cluster size only slightly and mainly changes how much cluster sizes vary. And finally, St has a nonlinear effect on all four summary statistics.